Algebra 2 — **Unit 3: Polynomial Functions**

Practice Test — Study Copy

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. Write each final answer in its box. Only what is inside a box is graded.

Part A — Vocabulary (8 pts)

Write the letter of the best description in the box next to each term.

1. Degree ☐
2. Leading coefficient ☐
3. End behavior ☐
4. Multiplicity of a zero ☐
5. Turning point ☐
6. Factor Theorem ☐
7. Rational Root Theorem ☐
8. Fundamental Theorem of Algebra ☐

(A) The number multiplying the highest-power term; its sign decides the right-hand arm.

(B) The greatest exponent on the variable once the polynomial is in standard form.

(C) How many times the factor $(x-r)$ appears — odd means the graph crosses at r, even means it touches.

(D) A point where the graph changes from increasing to decreasing, or the reverse.

(E) What the graph does at its far left and far right, as $x \to \pm\infty$.

(F) $(x-r)$ is a factor of $f(x)$ exactly when $f(r)=0$.

(G) Every rational zero of a polynomial with integer coefficients looks like $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.

(H) A polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity.

Part B — Multiple Choice (12 pts)

Write the letter of the best answer in its box.

☐ 1. The end behavior of $f(x) = -2x^4 + 3x^2 - 1$ is (A) both arms up (B) both arms down (C) left down, right up (D) left up, right down

☐ 2. $(2x-5)^2 =$ (A) $4x^2 - 25$ (B) $4x^2 + 25$ (C) $4x^2 - 20x + 25$ (D) $4x^2 - 10x + 25$

☐ 3. Factored completely, $x^3 + 27$ is (A) $(x+3)(x^2-3x+9)$ (B) $(x+3)(x^2+3x+9)$ (C) $(x+3)^3$ (D) prime

☐ 4. The remainder when $f(x) = x^3 - 4x^2 + 1$ is divided by $x-2$ is (A) -7 (B) -23 (C) 7 (D) 0

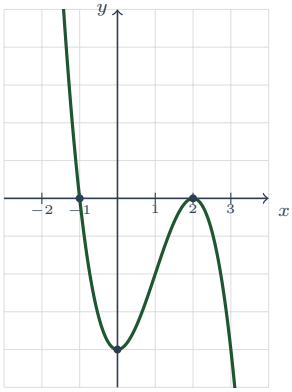
5. Which value is *not* a possible rational zero of $f(x) = 3x^3 + 2x^2 - x + 4$? (A) -2 (B) $\frac{4}{3}$ (C) $\frac{3}{4}$ (D) 1

6. A polynomial with *real* coefficients has degree 4 and zeros 2, -1, and $3i$. Its fourth zero must be (A) $-3i$ (B) $3i$ (C) -3 (D) $\frac{1}{3}$

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The polynomial function f is graphed below. Each gridline is one unit.



(a) Degree: even or odd? ____ Leading coefficient: positive or negative? ____

(b) As $x \to -\infty$, $f(x) \to$ ____; as $x \to +\infty$, $f(x) \to$ ____

(c) Zeros, each with its multiplicity, and whether the graph crosses or touches there: ____

(d) y-intercept: ____; number of turning points: ____

(e) Relative maximum: ____; relative minimum: ____;
absolute maximum or minimum: ____

(f) Interval(s) where f is increasing: ____; domain: ____; range: ____

2. **Operations.** Simplify. Write each answer in standard form.

(a) $(3x^3 - 5x + 2) - (x^3 + 4x^2 - 5)$

Answer: ____

(b) $(2x - 3)(x^2 + 4x - 1)$

Answer: ____

(c) For your answer to (b), give the degree ____, the leading coefficient ____, and the end behavior ____

3. **Factor completely.** Every factor must be prime over the integers.

(a) $2x^3 - 18x$ (b) $27x^3 - 8$ (c) $x^3 + 3x^2 - 4x - 12$

(a) [ ] **(b)** [ ] **(c)** [ ]

4. **Division and the Remainder Theorem.**

(a) Divide using *long division* and write the answer as $Q(x) + \dfrac{R}{D(x)}$: **Answer:** [ ]

$$(x^3 + 2x - 9) \div (x - 2)$$

(b) Use the *Remainder Theorem* to find the remainder when $g(x) = x^3 + 2x^2 - 5x + 6$ is divided by $x + 3$. Is $x + 3$ a factor of g? Explain in one sentence.

5. **Synthetic division.** You are told that $x = 3$ is a zero of

$$f(x) = x^3 - 2x^2 - 5x + 6.$$

Use synthetic division to divide out that zero, factor f completely, and list *all* of its zeros.

Factored: [ ] **Zeros:** [ ]

6. **Rational Root Theorem.** For $f(x) = 2x^3 - x^2 - 8x + 4$:

   (a) List every possible rational zero.

   Answer:

   (b) Find all real zeros of f, showing the test and the division you used.

   Zeros:

7. **Solve over the complex numbers.** $x^3 - 3x^2 + 4x - 12 = 0$

   (a) Before solving: how many solutions must this equation have, counted with multiplicity, and which theorem says so?

   (b) Solve. Give every solution.

   Solutions:

   (c) How many of your solutions are real, and how many x-intercepts does the graph of $y = x^3 - 3x^2 + 4x - 12$ therefore have?

8. **Build a polynomial from its zeros.** Write a polynomial function of *least degree* with *real* coefficients whose zeros are -1 and $2i$. Give your answer in factored form *and* in standard form.

   Factored: **Standard:**

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The graph plan.** Consider $f(x) = -2(x+2)(x-1)^2(x-3)$. Do *not* expand it. **(a)** State the degree and the leading coefficient, and explain how you got each one from the factored form. What end behavior do they force? **(b)** List the zeros with their multiplicities, and say whether the graph crosses or touches the x-axis at each. **(c)** Find the y-intercept. **(d)** Build a sign chart and state every interval on which $f(x) > 0$. **(e)** At most how many turning points can this graph have, and why?

2. **Modeling.** An open-top box is made from a 12 in by 6 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. **(a)** Write the volume $V(x)$ in factored form, then expand it into standard form. **(b)** State the feasible domain of the model and justify it from the sheet's dimensions. **(c)** Find $V(1)$ and state its units. **(d)** Check that $x = 6$ makes $V(x) = 0$. Explain why 6 is a genuine zero of the polynomial but not a possible cut length.